

Agroforestry Quantification

Greenhouse gas (GHG) removals attributable to the agroforestry component are quantified ex-ante using a modeled approach (QA1) in accordance with **Verra VM0042 v2.2** and the approved CDM methodological tool **AR-TOOL14: Estimation of carbon stocks and change in carbon stocks of trees and shrubs in A/R CDM project activities**^[1]^[2]. Agroforestry establishment on existing agricultural land is explicitly identified as an eligible project activity under VM0042 (Appendix 1), and biomass carbon stock changes from trees are permitted to be estimated using growth models and allometric equations under AR-TOOL14 (Sections 5 and 6).

The quantification follows a **carbon stock-change approach**, consistent with VM0042 Section 8 and AR-TOOL14 Section 4, whereby annual emission reductions are calculated as the year-to-year change in total biomass CO_{2e} stocks attributable to agroforestry establishment. The approach includes above-ground biomass, below-ground biomass, and retained dead biomass, and conservatively excludes soil organic carbon, litter, and harvested wood products.

The agroforestry activity is stratified by tree species based on the approved planting plan. For each species stratum, activity data include planted area (ha), planting density (trees/ha), and species-specific survival assumptions. All trees are assumed to be planted in Year 1 of the crediting period, forming a single age cohort per species. Tree biomass in the absence of the project is assumed to be zero, consistent with the absence of trees on agricultural land prior to project implementation.

Table X. Agroforestry tree species included in the project

<i>Common name</i>	<i>Scientific name</i>	<i>Functional type</i>	<i>Management context</i>
<i>Cashew</i>	<i>Anacardium occidentale</i>	Fast-growing fruit tree	Orchard / intercropped
<i>Shea</i>	<i>Vitellaria paradoxa</i>	Slow-growing savanna hardwood	Parkland agroforestry
<i>Mango</i>	<i>Mangifera indica</i>	Medium-fast fruit tree	Orchard / intercropped
<i>Moringa</i>	<i>Moringa oleifera</i>	Fast-growing softwood	Multipurpose agroforestry
<i>Dawadawa (Néré)</i>	<i>Parkia biglobosa</i>	Savanna hardwood	Parkland agroforestry

Tree growth is modeled ex-ante as a function of stand age using species-specific trajectories for diameter at breast height (DBH, cm) and height (m), consistent with AR-TOOL14 Section 5, which permits the use of growth models or yield tables for ex-ante estimation of biomass stocks^[2]. Where available, growth parameters are anchored to regionally relevant or species-specific literature from Northern Ghana, the Sudanian savanna, or comparable West African agro-ecological zones. Where such data are unavailable, conservative species-level or functional-group proxies are applied.

To represent realistic biological growth dynamics (establishment phase, juvenile acceleration, and maturity plateau), logistic growth functions are used as an interpolation mechanism between cited age–size anchor points and conservative maximum sizes. This approach is consistent with AR-TOOL14 guidance, provided that parameters are transparently documented and conservatively selected. Growth curves are used strictly to interpolate between literature-based anchor points and do not substitute for empirical data.

Above-ground biomass (AGB) per tree is estimated using species-specific allometric equations derived from peer-reviewed studies conducted in West Africa or comparable dry tropical systems. Depending on the species, equations use DBH alone or DBH in combination with height (and implicitly wood density), reflecting accepted allometric practice in tropical biomass estimation and consistent with AR-TOOL14 Section 6.

Table X. Allometric equations used to estimate above-ground biomass

Species	Allometric equation (AGB, kg d.m./tree)	Variables	Source
<i>Anacardium occidentale</i>	$AGB = 0.076 \times (DBH^2 \times H)^{0.927}$	DBH (cm), H (m)	Aabeyir et al. (2020)[^3]
<i>Vitellaria paradoxa</i>	$AGB = 0.084 \times DBH^{2.43}$	DBH (cm)	Dimobe et al. (2018)[^4]
<i>Mangifera indica</i>	Species-validated DBH-based equation	DBH (cm)	Oguntunde et al. (2011)[^5]
<i>Moringa oleifera</i>	Conservative pantropical equation	DBH (cm)	Orwa et al. (2009)[^6]
<i>Parkia biglobosa</i>	Savanna hardwood proxy equation	DBH (cm)	PROTA / CABI[^7]

Where no robust species-specific allometric equation is available, a conservative pantropical or functional-group proxy equation is applied, consistent with AR-TOOL14 provisions for data limitations.

Species-specific biomass estimation approaches

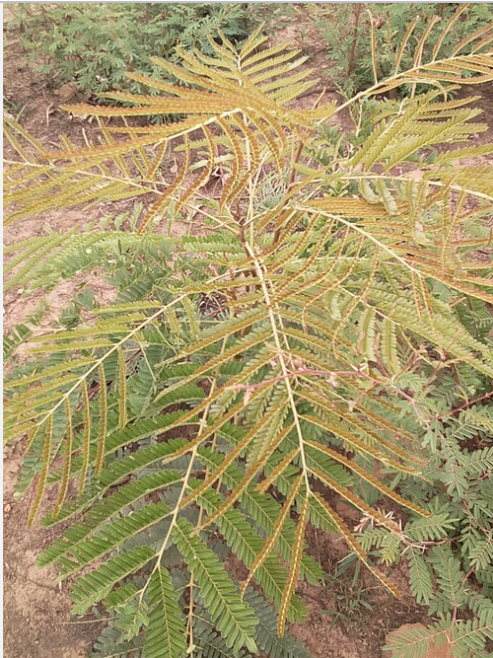
Tree	Rationale
 (Source: A cashew tree in Northern Ghana (2022), Wikimedia Commons)	Anacardium occidentale (Cashew): <i>Anacardium occidentale</i> is modeled as a fast-growing fruit tree established under orchard or intercropped agroforestry management, consistent with West African agricultural practice. DBH and height growth trajectories are anchored to plantation-based studies from West Africa and interpolated using a logistic growth function to represent establishment, juvenile growth, and maturity plateau. Above-ground biomass is estimated using a species-specific DBH–height allometric equation developed from Ghanaian woodland and plantation data, ensuring ecological relevance. Conservative parameter values and survival assumptions reflecting dryland agroforestry conditions

	are applied, ensuring realistic but non-overstated carbon accumulation.
 (Source: Shea tree from project site (2025) Gazelle)	Vitellaria paradoxa (Shea): <i>Vitellaria paradoxa</i> is a slow-growing, long-lived savanna hardwood typically managed under parkland agroforestry systems. Growth assumptions are intentionally conservative and based on Sudanian savanna studies, reflecting the species' slow diameter increment and delayed biomass accumulation. Above-ground biomass is estimated using a DBH-based allometric equation developed specifically for shea trees in West African parklands, avoiding reliance on height measurements that may introduce additional uncertainty. This approach prevents overestimation of early-period sequestration and is consistent with conservative ex-ante modeling under AR-TOOL14.
 (Source: Mango tree from project site (2025) Gazelle)	Mangifera indica (Mango): <i>Mangifera indica</i> is modeled as a medium-to-fast-growing fruit tree established under orchard or intercropped agroforestry management. Growth trajectories for DBH and height are anchored to plantation studies from Ghana and comparable agro-ecological contexts, with logistic interpolation used to reflect realistic juvenile growth and eventual maturity. Above-ground biomass is estimated using a species-validated DBH-based allometric approach consistent with published mango biomass studies. Conservative growth assumptions are applied to account for management practices such as pruning, ensuring that standing biomass is not overstated.



Source (Wiki commons)

Moringa oleifera (Moringa): *Moringa oleifera* is a fast-growing, low-wood-density species commonly managed as a multipurpose agroforestry tree. Given the limited availability of region-specific biomass allometry for mature moringa trees, growth trajectories are conservatively bounded using species profiles from the ICRAF Agroforestry Tree Database and comparable studies. Above-ground biomass is estimated using a conservative pantropical or functional-group proxy equation, consistent with AR-TOOL14 guidance where species-specific equations are unavailable. Conservative size bounds and exclusion of additional biomass pools ensure that carbon accumulation from moringa is not overstated despite rapid early growth.



(Source: Growing Dawadawa tree in northern Ghana Wikimedia)

Parkia biglobosa (Dawadawa / Néré): *Parkia biglobosa* is a characteristic savanna hardwood species managed under parkland agroforestry systems across West Africa. Height growth assumptions are anchored to regional species profiles and provenance studies, while diameter growth is modeled conservatively to reflect slow structural development relative to fruit trees. Due to the absence of robust species-specific biomass equations, above-ground biomass is estimated using a savanna hardwood proxy equation considered ecologically appropriate for the species' growth form. This approach aligns with AR-TOOL14 provisions for proxy use and ensures conservative estimation of biomass accumulation.

Ex-Ante quantification approach

Per-tree biomass estimates are converted to per-hectare biomass using the number of trees alive per hectare in each year, derived from species-specific cumulative survival functions. Below-ground biomass (BGB) is estimated using the root-to-shoot relationships prescribed in AR-TOOL14 Section 6, whereby BGB is calculated as a function of above-ground biomass density (t d.m./ha).

To avoid unrealistic instantaneous loss of carbon upon tree mortality, the model includes a retained dead biomass pool, consistent with the stock-change logic of AR-TOOL14. Reductions in live biomass between years are transferred to dead biomass and decay according to a first-order decay function using a conservative annual decay rate, representing gradual decomposition while maintaining temporal continuity of biomass carbon stocks.

Total biomass (live plus retained dead), expressed in tonnes dry matter per hectare (t d.m./ha), is converted to carbon using a default carbon fraction of 0.47 and subsequently converted to CO₂e using the molecular weight ratio 44/12, consistent with AR-TOOL14 default assumptions. Total agroforestry CO₂e stocks are calculated annually by aggregating per-hectare CO₂e across all species strata and multiplying by the corresponding planted area. Annual emission reductions are calculated as the year-to-year change in total CO₂e stocks attributable to agroforestry activities:

$$ER_t = CO_{2e_t} - CO_{2e_{t-1}}$$

Results are reported as total tonnes of CO₂e per year and, where required, normalized on a per-hectare basis. Conservative choices are applied throughout the quantification, including exclusion of soil carbon, use of regionally relevant growth data where available, conservative survival assumptions, and explicit modeling of mortality and dead biomass decay. All equations, parameters, and sources are transparently documented in the project calculation workbook.